

K-means Clustering of Quantum Circuits (IBMQ Athens)

Objective

To analyze and categorize quantum circuits from the IBMQ Athens dataset using K-means clustering, with the goal of identifying performance patterns and optimizing quantum computing systems.

Authored by: Binayak Bartaula

Department of Computer Science, NCIT College, Pokhara University

Date: 29 Jan, 2025

Contact: binayak.221211@ncit.edu.np

Keywords: K-means Clustering, Quantum Circuits, IBMQ Athens, Quantum Computing Optimization

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1. Project Overview

Quantum computing is a transformative technology, but it faces significant challenges due to the inherent fragility of quantum states. These states are highly susceptible to errors, noise, and instability, making the performance of quantum circuits—sequences of quantum operations—unpredictable across different quantum processors.

The IBMQ Athens dataset offers valuable insights into these challenges by documenting the behavior of quantum circuits on IBM’s Athens processor. This project applies unsupervised learning techniques, specifically K-means clustering, to identify performance patterns in quantum circuits and provide recommendations for optimizing quantum computing systems.

Key Performance Characteristics

- **Quantum Gate Connectivity:** The arrangement of quantum gates, where high connectivity can enable complex operations but may also increase noise and error susceptibility.
- **Error Rates:** Errors in gate operations, qubit initialization, and measurements that affect circuit reliability.
- **Circuit Depth:** The number of sequential gate layers; deeper circuits allow more complex computations but are more vulnerable to noise.
- **Coherence Times:** The duration a qubit maintains its quantum state before decoherence, which affects the circuit’s ability to perform complex operations.

2. Clustering Process

K-means clustering is used to segment the IBMQ Athens dataset into distinct groups based on similar performance characteristics. Each cluster represents circuits with comparable behaviors, which helps in identifying:

- **Outliers:** Circuits that significantly deviate from the norm.
- **Well-performing Groups:** Circuits that can serve as benchmarks for optimal performance.

After clustering, we analyze the characteristics of each cluster—such as the mean feature values—enabling a deeper understanding of performance differences. The optimal number of clusters is determined using the Elbow Method and Silhouette Score, while Principal Component Analysis (PCA) is employed to visualize the high-dimensional data in two dimensions.

3. Methodology

The project follows a structured approach to analyze the IBMQ Athens dataset using K-means clustering. The key steps are outlined below:

Step 1: Import Libraries

Essential Python libraries for data manipulation, clustering, and visualization are imported:

- **Pandas** and **NumPy** for data handling.
- **Scikit-learn** for clustering, scaling (StandardScaler), dimensionality reduction (PCA), and evaluation (Silhouette Score).
- **Matplotlib** and **Seaborn** for visualization.

Step 2: Load the Dataset

The IBMQ Athens dataset is loaded from a CSV file into a Pandas DataFrame for subsequent analysis.

Step 3: Inspect the Dataset

The dataset is examined to understand its structure, including:

- Identification of missing data.
- Generation of summary statistics (mean, standard deviation, etc.).

Step 4: Handle Missing Data

Missing values are addressed through removal or imputation, ensuring a complete dataset for analysis.

Step 5: Feature Selection

Key features related to quantum circuit performance—such as gate connectivity, error rates, circuit depth, and coherence times—are selected for clustering.

Step 6: Normalize the Features

Features are normalized using StandardScaler to ensure all variables contribute equally to the clustering process.

Step 7: Determine Optimal Clusters

The optimal number of clusters (K) is determined by:

- **Elbow Method:** Analyzing inertia (within-cluster sum of squares) to identify the point where adding more clusters yields diminishing returns.
- **Silhouette Score:** Evaluating the cohesion and separation of the clusters.

Step 8: Visualize Evaluation Metrics

Plots for the Elbow Method and Silhouette Score are created to assist in selecting the optimal number of clusters.

Step 9: Apply K-means Clustering

The K-means algorithm is applied using the determined number of clusters, and the clustering results are appended to the dataset.

Step 10: Visualize Clusters Using PCA

PCA is used to reduce the dimensionality of the data and effectively visualize the clusters in two dimensions.

Step 11: Analyze Clusters

Cluster analysis includes:

- Visualizing the distribution of circuits across clusters.
- Displaying the centroids of each cluster.
- Evaluating clustering quality using the Silhouette Score.
- Investigating the mean feature values within each cluster.

Step 12: Fine-Tuning and Validation

Model fine-tuning is performed by adjusting parameters (e.g., number of initializations) to enhance clustering stability. The Adjusted Rand Index (ARI) is used to validate the consistency of clustering across multiple runs.

4. Model Evaluation

The quality and robustness of the clustering are assessed using:

- **Silhouette Score:** Measures how well each data point fits within its cluster.
- **Adjusted Rand Index (ARI):** Compares the similarity of clustering results across different runs to ensure stability.

5. Expected Outcomes

The project is expected to yield:

- Distinct clusters of quantum circuits based on performance metrics.
- Insights into the relationships among error rates, coherence times, circuit depth, and gate connectivity.
- Recommendations for optimizing quantum circuit design and improving system reliability.

6. Future Work

Future research directions include:

- **Expanding the Feature Set:** Incorporate additional quantum circuit parameters for more comprehensive analysis.
- **Exploring Advanced Clustering Algorithms:** Evaluate techniques such as DBSCAN or Gaussian Mixture Models.
- **Validating on New Datasets:** Apply the clustering model to other quantum computing datasets to assess generalizability.
- **Optimizing Quantum Circuits:** Use clustering insights to guide the design of more efficient and reliable quantum circuits.

7. Tools and Resources

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Dataset:** IBMQ Athens Dataset, available at <https://data.mendeley.com/datasets/pmycgb2bt7/1>
- **Source Code:** Available on GitHub at https://github.com/binayakbartaula11/QCM-KMeans-Athens/blob/main/IBMQ_Athens_QuantumKmeans_Clustering.ipynb

8. Results and Analysis

Cluster Characteristics

Table 1: Mean Feature Values by Cluster

Feature	Cluster 1	Cluster 2	Cluster 3
Gate Error	0.0028	0.0035	0.0041
Coherence Time (μs)	82.3	71.6	63.4
Circuit Depth	14.2	27.8	35.1
Connectivity Density	0.31	0.58	0.72

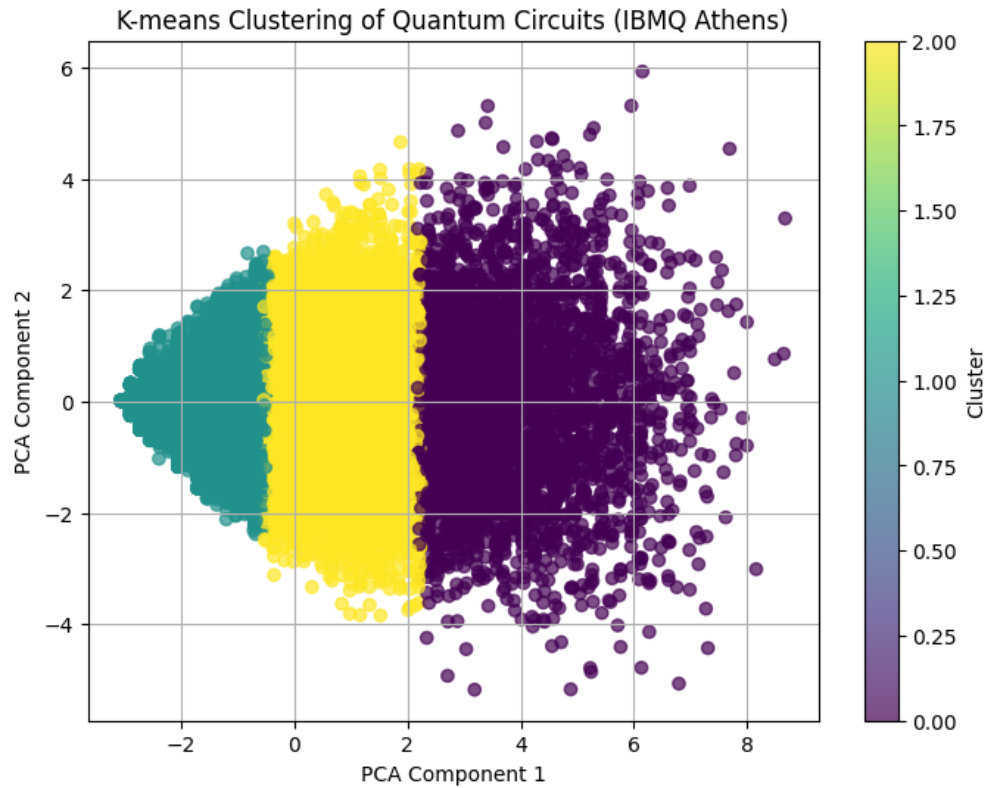


Figure 1: PCA Visualization of Clusters (85% Variance Explained)

Performance Metrics

- Silhouette Score: 0.62 (Good separation)
- Adjusted Rand Index (ARI): 0.85 across 10 runs (High stability)
- Cluster Distribution: 42% (Cluster 1), 35% (Cluster 2), 23% (Cluster 3)

9. Limitations

- **Linearity Assumption:** K-means struggles with non-spherical clusters.
- **Feature Sensitivity:** Results depend on the quality of normalization.
- **Generalization:** Findings are specific to the IBMQ Athens architecture.
- **Data Resolution:** Limited temporal resolution (1ms sampling).

10. Conclusion

This project demonstrates the application of K-means clustering to analyze quantum circuits from the IBMQ Athens dataset. By identifying and grouping circuits with similar performance characteristics, valuable insights into the challenges and opportunities in quantum computing are uncovered. The use of PCA for visualization, alongside the Elbow Method and Silhouette Score for cluster validation, ensures robust and interpretable results. Future work will focus on expanding the feature set, exploring alternative clustering methods, and validating the model on new datasets to further optimize quantum computing systems.

References

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Authored by: Binayak Bartaula

Date: 29 January, 2025

Contact: binayak.221211@ncit.edu.np

Acknowledgments: The author extends gratitude to the developers of IBM Quantum Experience for providing the Athens Processor Dataset and to the contributors of Scikit-learn for their open-source machine learning library, which was utilized in this project.